

Andrew Yuan

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Education

The Cooper Union for the Advancement of Science and Art, New York, NY Expected Graduation May 2026
Bachelor of Engineering in Electrical Engineering

- Double Minor in Mathematics and Computer Science
- GPA: 3.91/4.0 | Major GPA: 4.0/4.0 | Minor GPAs: 4.0/4.0
- **Awards and Honors:** Tau Beta Pi | Eta Kappa Nu | Jesse Sherman EE'33 Book Award – EE'26 Class Rank 1
Dean's List (All Semesters) | Half-Tuition Scholarship (2022-2025) | Full-Tuition Scholarship (2025-2026) |
Innovator's Award Scholarship
- **Relevant Coursework:** Linear Algebra, Probability Models and Stochastic Processes, Convex Optimization,
Discrete Mathematics, Abstract Algebra, Ordinary & Partial Differential Equations, Software Engineering & Large
Systems Design, Data Structures & Algorithms, Signals Processing, Finance

Technical Skills

Computer Languages: C, C++, Python, Java, SQL, MATLAB

Tools/Libraries: Git, NumPy, pandas, SciPy, Gurobi Spring Boot, JUnit, Docker, PostgreSQL, LTspice, $\LaTeX$

Experience

CAMP REU Research Intern, Cal Poly Pomona – Pomona, CA Summer 2025

- Studied kernel methods for Gaussian Processes and Support Vector Machines under Mentor Dr. Jimmy Risk
- Designed and optimized kernel functions in Python and R for use in applications such as time series, image processing, and financial data

Mathematics Tutor, The Cooper Union – New York, NY Fall 2024 – Present

- Provide drop-in assistance and individual tutoring for 5+ students in: Calculus I-III, Linear Algebra, Probability, and Differential Equations

Mathematics Grader, The Cooper Union – New York, NY Fall 2024 - Present

- Evaluate and provide feedback on assignments for 50+ students in Calculus I-II and Intro to Linear Algebra courses

Projects

CooperWall – Cooper Union Quote Forum, *ECE-366 Software Eng. Final Project* Spring 2025

- Collaborated with a three-member team to design and implement an interactive quote-sharing web application for the Cooper Union using a full-stack architecture
- Developed backend in Java using Spring Boot framework with PostgreSQL for persistent storage and Docker for containerized deployment.

Modeling Stat Optimization in RPG Game Mechanics Spring 2025

- Formulated a mixed-integer optimization model in Pyomo to maximize player combat damage, incorporating nonlinear game mechanics and stat scaling
- Utilized GLPK and Gurobi solvers to solve LP and QP relaxations; formulated and solved the dual problem to analyze sensitivity and characterize the optimal solution space
- Analyzed the feasibility region and active constraints using KKT conditions to assess the viability of the solution space

Extracurriculars

International Finance Student Association Student Chapter, *Secretary* Fall 2024 - Present

- Co-founded the first North American chapter of International Finance Student Association at Cooper Union, growing membership to 25+ active participants, surpassing the average club size of 10
- Coordinated numerous events and outreach with a budget of \$800 to ensure successful execution and financial sustainability over the course of the semester

US National Badminton Player 2012 - Present

- US Junior National Champion, Pan American Junior Champion, and US Adult National Finalist
- 5x US Junior National Team, 2025 FISU World University Games and 2022 World Junior Championships Team